# Improving Satellite Image Fusion via Generative Adversarial Training

Xin Luo, *Member, IEEE*, Xiaohua Tong, *Senior Member, IEEE*, and Zhongwen Hu, *Member, IEEE*

***Abstract*— The optical images acquired from satellite platforms are commonly multiresolution images, and converting multiresolution satellite images into full higher-resolution (HR) images has been a critical technique for improving the image quality. In this study, we introduced the generative adversarial network (GAN) and proposed a new fusion GAN (FusGAN) approach for solving the remote sensing image fusion problem. Specifically, we developed a new adversarial training strategy: 1) downscaled multiresolution images are adopted for generative network (G-Net) training, and 2) the discriminative network (D-Net) is used to adversarially train the G-Net by discriminating whether the original multiresolution images have been fused well enough. To further improve the capability of the network, we structured our G-Net with residual dense blocks by combining state-of-the-art residual and dense connection ideas. Our proposed FusGAN approach is evaluated both visually and quantitatively on Sentinel-2 and Landsat Operational Land Imager (OLI) multiresolution images. As demonstrated by the results, the proposed FusGAN approach outperforms the selected benchmark methods and both perfectly preserves spectral information and reconstructs spatial information in image fusion. Considering the common resolution disparities among intra- and intersatellite images, the proposed FusGAN approach can contribute to the quality improvement of satellite images and thus improve remote sensing applications.**

***Index Terms*— Deep learning, generative adversarial networks (GANs), Landsat 8, remote sensing image fusion, residual dense blocks, Sentinel-2.**

## I. Introduction

REMOTE sensing images provide essential knowledge for understanding the dynamics of Earth, and they have been widely used for land management and planning [1], ecosystem studies [2], environmental change monitoring [3], and biodiversity conservation [4]. Information on the surface of Earth can be captured from satellite images with a specific ground sampling distance (also referred to as the spatial resolution), and images with a higher spatial are preferred in some remote sensing applications [5], [6]. Nevertheless, due to the limitations of the payload, cost, complexity, and imaging system of satellites, the acquired satellite images commonly vary in spatial resolution, that is, a combination of higher resolution (HR) and lower-resolution (LR) spectral bands is usually acquired for Earth observation (e.g., QuickBird, WorldView-3, Sentinel-2, and Landsat-8). Generally, the limited spectral information of the HR image hinders accurate discrimination between spectrally similar object in the image, and the limited spatial information of the LR image restricts fine extraction of the small objects in the image ([7], [8]), to further improve the satellite image quality, therefore, fusing multiresolution images into full high-spatial resolution images has been studied.

There are two main approaches for remote sensing image fusion: component substitution (CS) [9] and multiresolution analysis (MRA) [10]. The CS approach is based on the projection of the lower-spatial resolution image into a new domain and the substitution of components with a finer-spatial resolution band. Classical CS methods include the intensity-hue-saturation (IHS) method [11]–[13], Brovey transformation (BT) [14], Gram–Schmidt transformation [15], principal component analysis (PCA) [9], [13], [16], [17], and partial replacement adaptive CS (PRACS) [18]. The recently proposed methods belonging to this class can be found in the literature and include those by Xu *et al.* [19] and Kang *et al.* [20]. The MRA approach is based on the extraction of spatial details from the fine-spatial resolution image and their injection into the higher-frequency domain of the coarse-spatial resolution image. Spatial details can be extracted in different ways, for example, decimated or undecimated wavelet transforms [21]–[25], Laplacian pyramids [26]–[28], and other nonseparable transforms, such as contourlets [17], [29], [30] and curvelets [31], [32].

Certain approaches that do not fall in either the CS or MRA category also occur in the literature. For example, kriging-based geostatistical approaches such as kriging with external drift (KED) [33], downscaling cokriging (DSCK) [34]–[36], and the area-to-point regression kriging (ATPRK) approach [37], [38] have also been developed for satellite imagery fusion. The compressive sensing theory, which was first applied by Li and Yang [39] in pansharpening work,

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Xin Luo is with the MNR Key Laboratory for Geo-Environmental Monitoring of Great Bay Area & Guangdong Key Laboratory of Urban Informatics & Shenzhen Key Laboratory of Spatial Smart Sensing and Services, Shenzhen University, Shenzhen 518060, China, and also with the College of Life Sciences and Oceanography, Shenzhen University, Shenzhen 518060, China (e-mail: xinluo_xin@szu.edu.cn).

Xiaohua Tong is with the College of Surveying and Geo-informatics, Tongji University, Shanghai 200092, China (e-mail: xhtong@tongji.edu.cn).

Zhongwen Hu is with the MNR Key Laboratory for Geo-Environmental Monitoring of Great Bay Area & Guangdong Key Laboratory of Urban Informatics & Shenzhen Key Laboratory of Spatial Smart Sensing and Services, Shenzhen University, Shenzhen 518060, China.

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was further improved by Jiang *et al.* [40], Jiang *et al.* [41], Wang *et al.* [42], and Ghahremani *et al.* [43]. Recently, due to the rapid development of deep learning technology, deep learning-based methods have also been introduced to solve the problem of remote sensing imagery fusion [44]–[52]. Specifically, Masi *et al.* [44] adopted a simple and effective three-layer convolutional neural network (ConvNet) for satellite image fusion, and with augmentation of the input with several spectral index images, the proposed ConvNet achieved significant improvement in image fusion. This three-layer ConvNet was further improved by Scarpa *et al.* [46] using residual learning and L1 loss. In addition, Scarpa *et al.* [46] introduced a target-adaptive tuning phase, which was used to solve the problem of insufficient training data, thereby improving the applicability of the newly proposed ConvNet. ConvNet has also been designed to enhance image fusion. Wei *et al.* [48] and Palsson *et al.* [53] created a very deep ConvNet with the residual learning architecture, and Lanaras *et al.* [54] designed DSen2 with multiple ResBlock modules. Through full training with globally sampled data, the pre-trained DSen2 was used to fuse regional Sentinel-2 imagery without additional retraining. In addition, GAN theory [55] are also introduced to improve the multiresolution fusion of remote sensing image, mostly for the high-resolution images [56]–[58]. As demonstrated by the literature, the recently developed deep learning-based methods perform better than the non-deep learning-based methods in remote sensing image fusion; moreover, they still have great potential to be further refined.

In this study, to further exploit the potential of the deep learning-based image fusion method, we designed a new GAN-based deep learning model to solve the multiresolution fusion problem in satellite images, particularly for moderate-resolution images. Generally, we contribute to current image fusion research in three main aspects.

1) A new generative adversarial training framework is designed to suppress the errors caused by the scale invariance assumption, thus improving deep learning-based image fusion. Through adversarial training using pairwise fused images and pair-wise real images, the GAN-based framework performs favorably against the ConvNet-based framework in satellite image fusion.
2) The use of state-of-the-art residual dense blocks was the first attempt at solving the satellite image fusion problem. Through experimental analysis, residual dense blocks demonstrated significant improvements in satellite image fusion when compared with the traditional convolutional blocks and advanced residual blocks.
3) A new FusGAN model is structured for the satellite image fusion using auto-generated degraded pair-wise multiresolution images. To extend the general applicability, this method is designed for satellite images that vary in resolution and exhibit different combinations of HR and LR spectral bands.

The remainder of this article is organized as follows. In Section II, we describe the details of the proposed method. Then, we present the test data and related image fusion results in Sections III and IV, respectively. Finally, discussions and conclusions are provided in Sections V and VI, respectively.

## II. Method

The deep learning-based method captures the transfer relation between the multiresolution and full HR images. In this study, based on the GAN theory, we developed a new generative adversarial training framework for remote sensing image fusion. Generally, the generative adversarial training framework contains three steps (see Fig.1):

*Step 1:* Generative network (G-Net) training based on remote sensing images with a degraded resolution. In this step, due to the lack of a reference image, we downscale multiresolution images with an image degradation algorithm, and then, the original LR image can be adopted as a reference for G-Net training.

*Step 2:* Application of the G-Net to original-resolution image fusion. The G-Net, which is trained on degraded images, is applied to original-resolution image fusion; accordingly, superresolution LR images can be obtained.

*Step 3:* Adversarial discriminative network (D-Net) training by using fused and original pairwise images. Specifically, a pairwise image consists of a downscaled superresolved LR image, and the original HR image is regarded as the fake sample, while the pairwise original multiresolution image is regarded as the real sample.

D-Net aims to correctly discriminate fused pairwise images from original pairwise images, while the G-Net aims to reconstruct real enough pairwise images to challenge the D-Net. Through consecutive training loops between the G-Net and D-Net, the G-Net can accurately capture the transfer relation from the multiresolution images to the full HR images.

### A. G-Net With Residual Dense Blocks

The G-Net aims to convert multiresolution LR and HR images into superresolution LR images. We use $S_{\mathrm{LR}}$ and $S_{\mathrm{HR}}$ to represent LR and HR images, respectively; then, the objective of the G-Net is to obtain superresolution $S_{\mathrm{LR}}$ images and hence, together with $S_{\mathrm{HR}}$ images, to produce full HR images. The transfer relation function can be described as $S_{\mathrm{LR}}^{\mathrm{SR}} = G(S_{\mathrm{LR}}, S_{\mathrm{HR}})$. To achieve the goal of high-fidelity superresolution, we adopted residual dense blocks based on the idea of ESRGAN [63] to build the G-Net. Specifically, the residual dense blocks in ESRGAN remove the batch normalization layers from the original residual blocks in the ResNet [59] and moreover, dense blocks [60] are used in the main path similar to Zhang *et al.* [61], and the network thus benefits more from the dense connections. The use of residual dense blocks significantly improved the performance of ESRGAN in generic image superresolution, and therefore, we adopted them for G-Net structuring in our study. The residual dense blocks and of the G-Net are shown in Fig. 2. Notably, in the first concatenation operation of the residual dense block, we concatenate the input feature maps $x_l$ and

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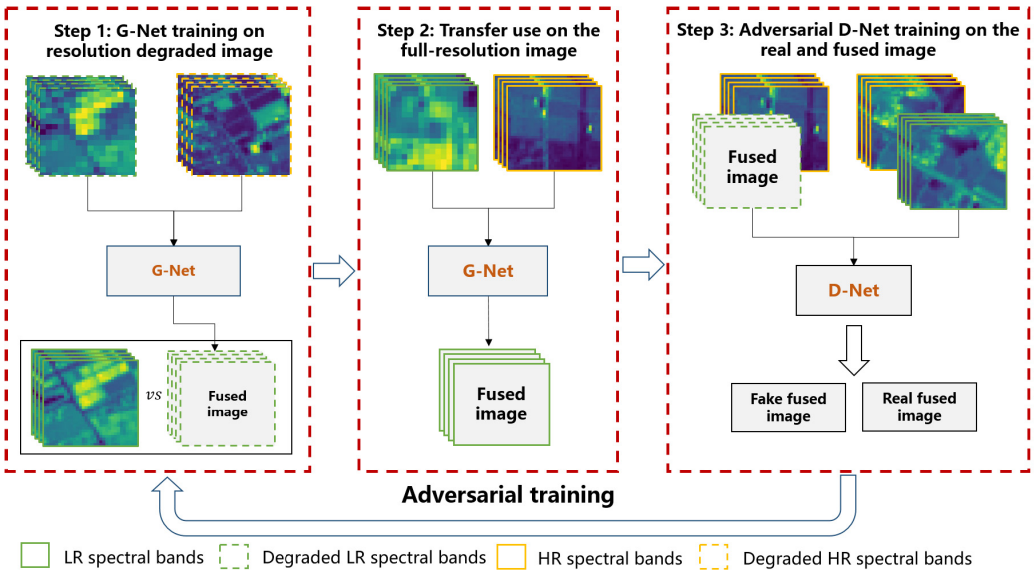


Fig. 1. New framework of GAN-based satellite image fusion. LR, HR, and SR indicate LR, HR, and superresolution images, respectively.

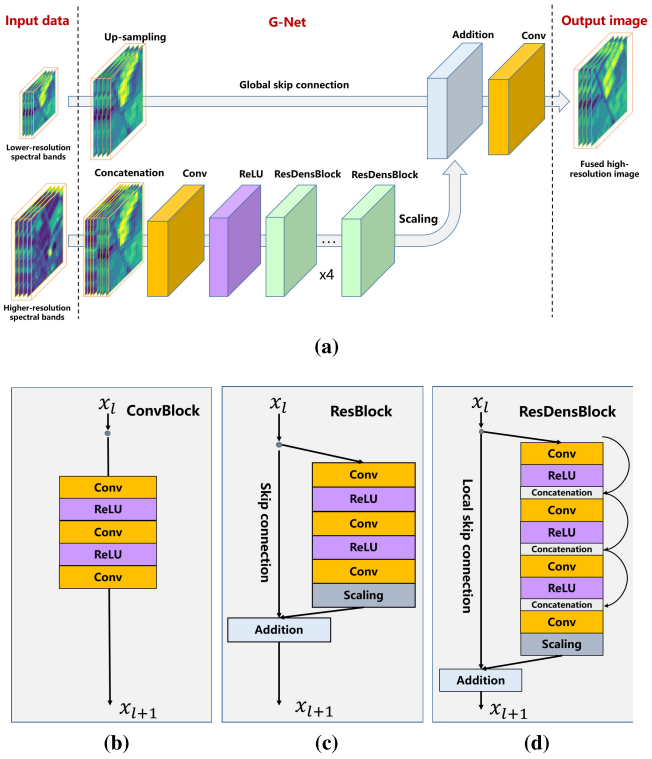


Fig. 2. Architecture of the G-Net with residual dense blocks. (a) Architecture of the structured G-Net. (b) Traditional ConvBlock. (c) Advanced ResBlock. (d) State-of-the-art ResDensBlock.

the feature maps derived by the followed convolution as well as the ReLU activation operations. In the subsequent concatenation operations, the ReLU activation map and the previous concatenated feature maps are concatenated [Fig. 2(d)].

The ultimate goal of the G-Net is to generate high-fidelity superresolution images. We use the superresolution loss function to quantify the error between the superresolution and reference images. Then, the G-Net training procedure becomes a minimization process of the superresolution loss function. We follow the studies by Lim *et al.* [62] and Wang *et al.* [63] and train the G-Net with the L1-norm loss function

$$\begin{aligned}\text{Loss}^{\text{SR}} &= E_{S_{\text{SR}},S_{\text{HR}},S_{\text{LR}}^{\text{Ref}}}\left\|S_{\text{LR}}^{\text{Ref}} - G(S_{\text{LR}},S_{\text{HR}})\right\|_1 \\ &= \frac{1}{\text{WH}}\sum_{x=1}^{W}\sum_{y=1}^{H}\left\|S_{\text{LR}}^{\text{Ref}} - S_{\text{LR}}^{\text{SR}}\right\|_1 \end{aligned} \tag{1}$$

where $S_{\text{LR}}^{\text{SR}}$ and $S_{\text{LR}}^{\text{Ref}}$ are the superresolved LR and reference images, respectively, and $W$ and $H$ are the width and height, respectively, of the superresolved image.

### *B. Markovian D-Net With the Fusion Loss Function*

Superresolved LR and original HR images can be obtained by using the G-Net; accordingly, we designed the D-Net for discriminating whether the pairwise image is fused well enough by the adversarial method. Specifically, if the pairwise image contains a component that is generated by the G-Net, the discrimination result is expected to be fake; otherwise, the discrimination result is expected to be real, and the discrimination function can be expressed as Real/Fake $= D(S_{\text{LR}}/S_{\text{LR}}^{\text{SR}}, S_{\text{HR}})$. During adversarial training, the G-Net is forced to obtain a better fusion result to fool the D-Net, and the G-Net is required to constantly improve its ability to correctly discriminate fused pairwise images. We can formulate the discrimination process as a classification problem, and the fusion loss function can then be defined with the cross-entropy based on the probabilities derived by the D-Net across all the real and fake pairwise training samples as follows:

$$\begin{aligned}\text{Loss}^{Fus} = &-E_{S_{\text{LR}},S_{\text{HR}}}\log(1 - D(G(S_{\text{LR}},S_{\text{HR}}),S_{\text{HR}})) \\ &- E_{S_{\text{LR}},S_{\text{HR}}^{\text{Dr}}}\log D\left(S_{\text{LR}}^{\text{SR}},S_{\text{HR}}\right).\end{aligned} \tag{2}$$

Our D-Net is built on the Markovian PatchGAN idea [64], and such a D-Net effectively models an image as a Markov random field, assuming independence between pixels separated by a specific patch diameter. The Markovian generator only

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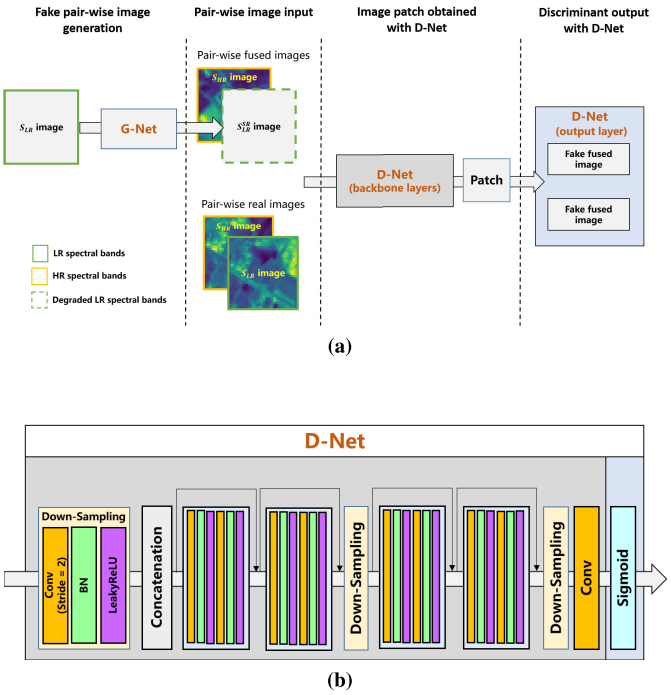


Fig. 3. Architecture of the D-Net, and its role in adversarial training. (a) Workflow of adversarial training by D-Net. (b) Details of the D-Net structure.

penalizes the image structure at the scale of the image patch and is therefore capable of capturing more detailed structural information of the image. As pairwise image discrimination mainly depends on the detailed structural information of the local image region, Markovian design is particularly suitable for D-Net structuring in our study. Detailed information of the structured D-Net and its role in adversarial training are shown in Fig. 3.

### C. Adversarial Training and Inference

Following Goodfellow *et al.* [55], the D-Net is trained to distinguish fused pairwise images from original pairwise images, while the G-Net is trained with the goal of fooling the D-Net. With for-loops in adversarial training, the G-Net can learn to create high-fidelity superresolved images that are difficult to distinguish by the D-Net. We define the adversarial loss as

$$\text{Loss}^{\text{Adv}} = E_{S_{\text{LR}},S_{\text{HR}}} \log(1 - D(G(S_{\text{LR}}, S_{\text{HR}}), S_{\text{HR}})) \tag{3}$$

where $D(G(S_{\text{LR}}, S_{\text{HR}}), S_{\text{HR}})$ is the probability that the generated pairwise image $[G(S_{\text{LR}}, S_{\text{HR}}), S_{\text{HR}}]$ is a real pairwise image. To perform adversarial training against the D-Net, G-Net training should optimize both $\text{Loss}^{\text{SR}}$ and $\text{Loss}^{\text{Adv}}$, that is,

$$G_{\text{Loss}} = \text{Loss}^{\text{Adv}} + \lambda \text{Loss}^{\text{SR}} \tag{4}$$

where $\lambda$ controls the importance of $\text{Loss}^{\text{SR}}$ and $\text{Loss}^{\text{Adv}}$, and we constantly set $\lambda$ to 1000 in our study. To more easily understand the adversarial training between the G-Net and D-Net, we divide the training procedure into the following two steps.

**Core Algorithms for Generative Adversarial Training**

Step 1: train the D-Net by:

$$\min_{D} -E_{S_{\text{LR}},S_{\text{HR}}^{\text{Dr}}} \log D\left(S_{\text{LR}}, S_{\text{HR}}^{\text{Dr}}\right) - E_{S_{\text{LR}},S_{\text{HR}}} \log(1 - D(G(S_{\text{LR}}, S_{\text{HR}}), S_{\text{HR}}))$$

Step 2: train the G-Net by:

$$\min_{G} \; E_{S_{\text{LR}},S_{\text{HR}},S_{\text{LR}}^{\text{Ref}}} \left\| S_{\text{LR}}^{\text{Ref}} - G(S_{\text{LR}}, S_{\text{HR}}) \right\|_1 + E_{S_{\text{LR}},S_{\text{HR}}} \log(1 - D(G(S_{\text{LR}}, S_{\text{HR}}), S_{\text{HR}})).$$

As suggested in the original GAN article [55], rather than training the G-Net to minimize $\log(1 - D(G(S_{\text{LR}}, S_{\text{HR}}), S_{\text{HR}}))$, we can train the G-Net to maximize $\log(D(G(S_{\text{LR}}, S_{\text{HR}}), S_{\text{HR}}))$; therefore, the ultimate objective function can be simply expressed as an adversarial minimum-maximum problem

$$\begin{aligned} G^* = \arg\min_{G}\max_{D} & E_{S_{\text{LR}},S_{\text{HR}}^{\text{Dr}}} \log D\left(S_{\text{LR}}, S_{\text{HR}}^{\text{Dr}}\right) \\ & - E_{S_{\text{LR}},S_{\text{HR}}} \log(D(G(S_{\text{LR}}, S_{\text{HR}}), S_{\text{HR}})) \\ & + E_{S_{\text{LR}},S_{\text{HR}},S_{\text{LR}}^{\text{Ref}}} \left\| S_{\text{LR}}^{\text{Ref}} - G(S_{\text{LR}}, S_{\text{HR}}) \right\|_1. \end{aligned} \tag{5}$$

### D. Evaluation Metrics

Evaluation with a visual check and accuracy quantification are both performed in our study. Particularly, four metrics of the spectral angle mapper (SAM) (in degrees), signal-to-reconstruction error (SRE), root mean square error (RMSE), and structural similarity (SSIM) are adopted for the quantitative accuracy assessment, and the four metrics can be calculated as follows:

$$\text{SAM} = \arccos\left(\frac{x_p^T \hat{x}_p}{\|x_p\| \|\hat{x}_p\|}\right) \cdot \frac{180}{\pi} \tag{6}$$

$$\text{SRE} = 10\log_{10}\left(\frac{\|\hat{x}\|^2}{\|\hat{x} - x\|^2}\right) \tag{7}$$

$$\text{MAE} = \sqrt{\frac{1}{n}\sum(\hat{x} - x)} \tag{8}$$

$$\text{SSIM} = \frac{2(u_x u_{\hat{x}} + c_1)(2\sigma_{x\hat{x}} + c_2)}{\left(u_x^2 + u_{\hat{x}}^2 + c_1\right)\left(\sigma_x^2 + \sigma_{\hat{x}}^2 + c_2\right)} \tag{9}$$

where $x_p$ and $\hat{x}_p$ are the spectral vectors of the pixels in the fused and reference images, respectively, at location $p$, and $u_x$, $\sigma_x$, $u_{\hat{x}}$ and $\sigma_{\hat{x}}$ represent the mean and standard deviation of the fused and reference images, respectively, in a local image window centered at pixel $x$. Moreover, $\sigma_{x\hat{x}}$ is the covariance between the fused and reference images, and $c_1$ and $c_2$ are constants. The size of the local image window is set to 11 in our study.

## III. Experiments and Method Evaluation

### A. Test Data: Landsat 8 and Sentinel-2 Images

Landsat 8 and Sentinel-2 images are selected for method testing in this study. The Landsat series of satellites supports the longest temporal (over 40 years) space-based Earth observations [65], which has resulted in a large amount of archived

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TABLE I

CHARACTERISTICS OF THE SPECTRAL BANDS OF THE SELECTED MULTI-RESOLUTION LANDSAT OLI AND SENTINEL-2 IMAGES. SOME SPECTRAL BANDS FOR SPECIFIC USES HAVE BEEN REMOVED

| **Landsat OLI** | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Band number | B8 | B1 | B2 | B3 | B4 | B5 | B6 | B7 | | |
| Central wavelength (nm) | 590 | 440 | 480 | 560 | 655 | 865 | 1610 | 2200 | | |
| Bandwidth (nm) | 180 | 20 | 60 | 60 | 30 | 30 | 80 | 180 | | |
| Spatial resolution (m) | **15** | **30** | **30** | **30** | **30** | **30** | **30** | **30** | | |
| **Sentinel-2** | | | | | | | | | | |
| Band number | B2 | B3 | B4 | B8 | B5 | B6 | B7 | B8a | B11 | B12 |
| Central wavelength (nm) | 490 | 560 | 665 | 842 | 705 | 740 | 783 | 865 | 1610 | 2190 |
| Bandwidth (nm) | 65 | 35 | 30 | 115 | 15 | 15 | 20 | 20 | 90 | 180 |
| Spatial resolution (m) | **10** | **10** | **10** | **10** | **20** | **20** | **20** | **20** | **20** | **20** |

remote sensing data. The Landsat 8 satellite has a two-sensor payload [66]; one sensor is the OLI, which collects eight spectral bands with a 30-m spatial resolution and one panchromatic band with a spatial resolution of 15 m, and the other sensor is the Thermal Infrared Sensor (TIRS), which collects two thermal bands with a resolution of 100 m. To supplement the systematic global acquisition of HR multispectral imagery, the European Space Agency (ESA) launched Sentinel-2A and Sentinel-2B paired satellites on June 23, 2015 and March 7, 2017 [67], respectively. Unlike most satellite images that only have one finer-resolution panchromatic band, the Sentinel-2 Multispectral Instrument (MSI) is designed to acquire multiresolution multispectral images, that is, the acquired images contain four spectral bands at 10 m, six spectral bands at 20 m and three spectral bands at 60 m. Due to the atmospheric corrections of the Sentinel-2 60-m bands and cloud screening of the Landsat 8 cirrus band [67], we removed the cirrus band of the Landsat OLI images and the 60-m spectral bands of the Sentinel-2 images, and the remaining multiresolution Landsat OLI and Sentinel-2 images are used for the image fusion tests. More details about the selected images are shown in Table I below. Notably, the overlap modes between the multiresolution Landsat 8 and Sentinel-2 images are different. The Landsat image overlap mode has boundary shifts between the LR and HR bands (Fig. 4). In this study, we unify the overlap mode to the Sentinel-2 mode; specifically, we crop the boundary region of the $\times 4$ upsampled LR image beyond the edges of the HR image, and then the $4 \times 4$ aggregation algorithm is applied to restore the resolution of the original LR image (see in Fig. 4). In addition, for the training data of pair-wise multiresolution images, we follow the Wald protocol [54], [68], which assumes scale invariance, such that the downscaled HR and LR images pair can be used to learn the parameters of the fusion of the original HR and LR images pair. Specifically, we downscale multiresolution images by automatically applying a Gaussian blur filter and a downsampling step to the original images; then, the original images are used as ground truth for the model training.

### *B. Experimental Results and Analysis*

Full HR Landsat OLI and Sentinel-2 images can be obtained by implementing the proposed FusGAN approach. To evaluate the superiority of the new method, five existing image fusion methods are selected. The methods to be compared include two classical methods, i.e., Gram-Schmidt pansharpening (G-S, [69] and the nearest-neighbor diffusion-based pansharpening algorithm (NNDiffuse, [70]), two state-of-the-art methods, i.e., the morphological operators-based fusion method (MF, [71]), the ATPRK method [37], [38], and the newly proposed deep learning-based DSen2 method [54]. As many of the existing methods (i.e., G-S, NNDiffuse, and MF) are applicable only to multiresolution combinations of one HR panchromatic band and multiple LR spectral bands, to enable the successful fusion of Sentinel-2 images, we simulated an HR panchromatic band through $\text{Pan} = \int [R, G, B, \text{Nir}]$, and the simulated panchromatic band was then applied in Sentinel-2 image fusion. On the other hand, DSen2 is tailored for Sentinel-2 image fusion; therefore, to enable suitable fusion of Landsat images by using DSen2, we altered the input layer of the original DSen2 network and kept the other set the same. The parameters for FusGAN training have been previously determined in our study. Specifically, for the parameter $\lambda$ in (4), we follow the research by Ledig *et al.* [72] and set $\lambda$ to 1000. For the key parameters of the number of residual dense blocks, the patch size of the training data, and the learning rate, we searched the appropriate parameters within the scope of $[2, 3, 4, 5, 6]$, $[32, 48, 64]$, and $[0.01, 0.001, 0.0001, 0.00001]$, respectively. Accordingly, the convolution kernel size, the number, the stride, the batch size, the number of image patches, the training epochs, the number of residual dense blocks, the patch size, and the learning rate are finally determined as $3 \times 3$, 32, 1, 4, 1000, 200, 4, 48, and 0.0001 in our study.

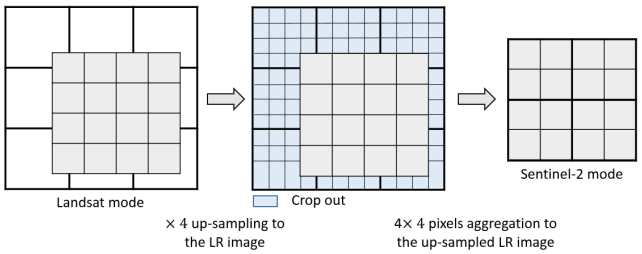


Fig. 4. Overlap mode unification of the Landsat mode to the Sentinel-2 mode.

*1) Quantitative Accuracy Assessment:* Quantitative evaluation of fused images is a difficult issue due to the lack of

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TABLE II

PERFORMANCE INDICATORS OF THE SIMULATED FULL-SPECTRAL (REDUCED SPATIAL RESOLUTION) AND FULL-SPATIAL RESOLUTION (REDUCED SPECTRAL BANDS) IMAGES. THE BEST RESULTS ARE IN BOLD FONT

| | Full-spectral assessment | | | | Full-spatial resolution assessment | | | |
|---|---|---|---|---|---|---|---|---|
| | MAE | SAM | SSIM | SRE | MAE | SAM | SSIM | SRE |
| G-S | 0.0292 | 5.535 | 0.8329 | 18.524 | 0.0164 | 3.923 | 0.9066 | 20.827 |
| NNDiffuse | 0.0134 | 2.674 | 0.9273 | 25.607 | 0.0125 | 2.6859 | 0.946 | 22.06 |
| ATPRK | 0.0128 | 3.242 | 0.933 | 28.293 | 0.0089 | 3.0649 | 0.9538 | 30.452 |
| MF | 0.0165 | 3.131 | 0.8857 | 26.584 | 0.011 | 2.923 | 0.9214 | 28.995 |
| DSen2 | 0.0064 | 1.9233 | 0.9763 | 32.805 | 0.0053 | 2.2795 | 0.9773 | 32.534 |
| FusGAN | **0.004** | **1.321** | **0.9905** | **36.863** | **0.0041** | **1.772** | **0.9839** | **35.473** |

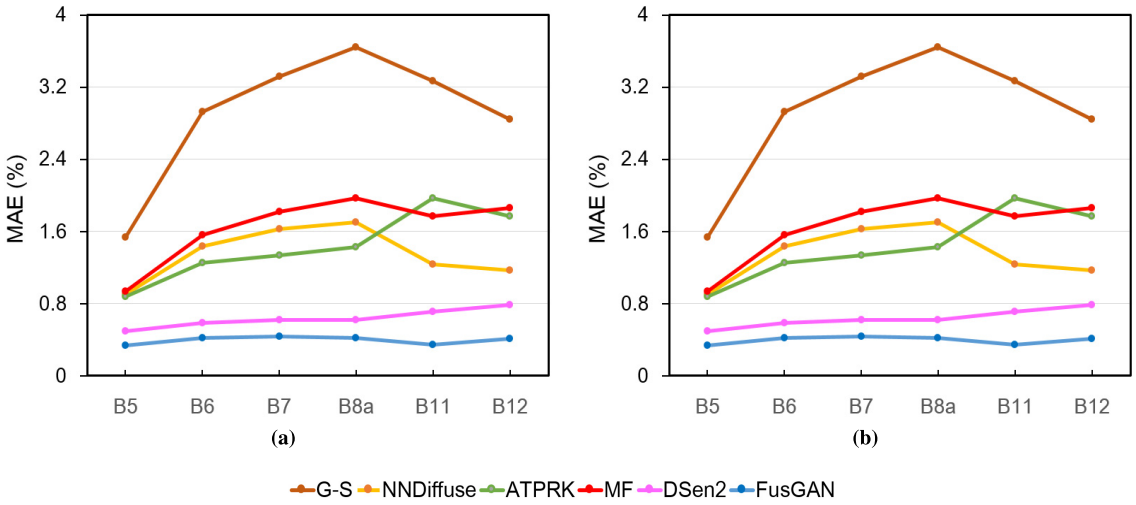


Fig. 5. Illustration of the MAE value for each band of the fused images. (a) Full-spectral assessment. (b) Full-spatial resolution assessment.

reference superresolved images. By following Wald's protocol [68], many researchers [38], [54], [73], [74] quantitatively evaluate images with a reduced spatial resolution, that is, reduced-spatial resolution images are generated by applying a Gaussian blur filter and a downsampling step to the original images, and the original images are then used as references for validating fused downsampled images. The reduced-spatial resolution image retains all the spectral information; however, a problem still exists since the image is obtained under the premise of scale invariance, which is not completely rigorous in the learning-based superresolved image approach. For that reason, the method has to be supplemented by full-resolution image assessment, and sufficient evidence must be provided for image fusion validation. In this study, we adopt simulated Sentinel-2 images for the quantitative accuracy assessment. Specifically, 1) we apply a Gaussian filter with radius $=3$ and sigma $=0.5$ and a downsampling factor of $1/2$ to the original Sentinel-2 images. Then, the obtained images with a reduced resolution but fully preserved spectral bands are used for full-spectral assessment. Moreover, 2) in addition to full-spectral assessment, we integrate the Sentinel-2 10-m RGB spectral bands into a panchromatic band and apply the same downscaling algorithm as that in the full-spectral band simulation to the Sentinel-2 10-m spectral bands. Then, simulated images with the original spatial resolution are obtained for full-resolution assessment. The parameters for FusGAN training have been previously determined in our study, that is, the batch size, number of image patches, training epochs, residual dense blocks and filters in each convolutional layer, patch size, and learning rates of the G-Net and D-Net are 4, 1000, 200, 4, 32, 48, 0.0001 and 0.002, respectively. We first calculated the overall accuracy of the fused images by using the indicators of MAE, SAM SSIM and SRE. As Table II indicates, the proposed FusGAN approach performs the best in both full-spectral image fusion and full-spatial resolution image fusion. In full-spectral band assessment, the FusGAN approach achieved the highest SSIM and SRE values of 0.9905 and 36.863, respectively, and the lowest MAE and SAM values of 0.0040 and 1.321, respectively. In full-spatial resolution assessment, the FusGAN approach achieved the highest SSIM and SRE values of 0.9839 and 35.473, respectively, and the lowest MAE and SAM values of 0.0041 and 1.772, respectively. We evaluated each spectral band of the fused images by using the error indicator of MAE. As demonstrated in Fig. 5, the proposed FusGAN approach attained the lowest fusion errors of all the spectral bands in both full-spectral image fusion and full-spatial resolution image fusion. In particular, the HR band of the simulated full-resolution images only covers the visible spectrum, which leads to an accuracy decrease in the near-infrared band of the fused images in full-resolution image fusion. Based on full-spectral resolution image fusion assessment, the near-infrared

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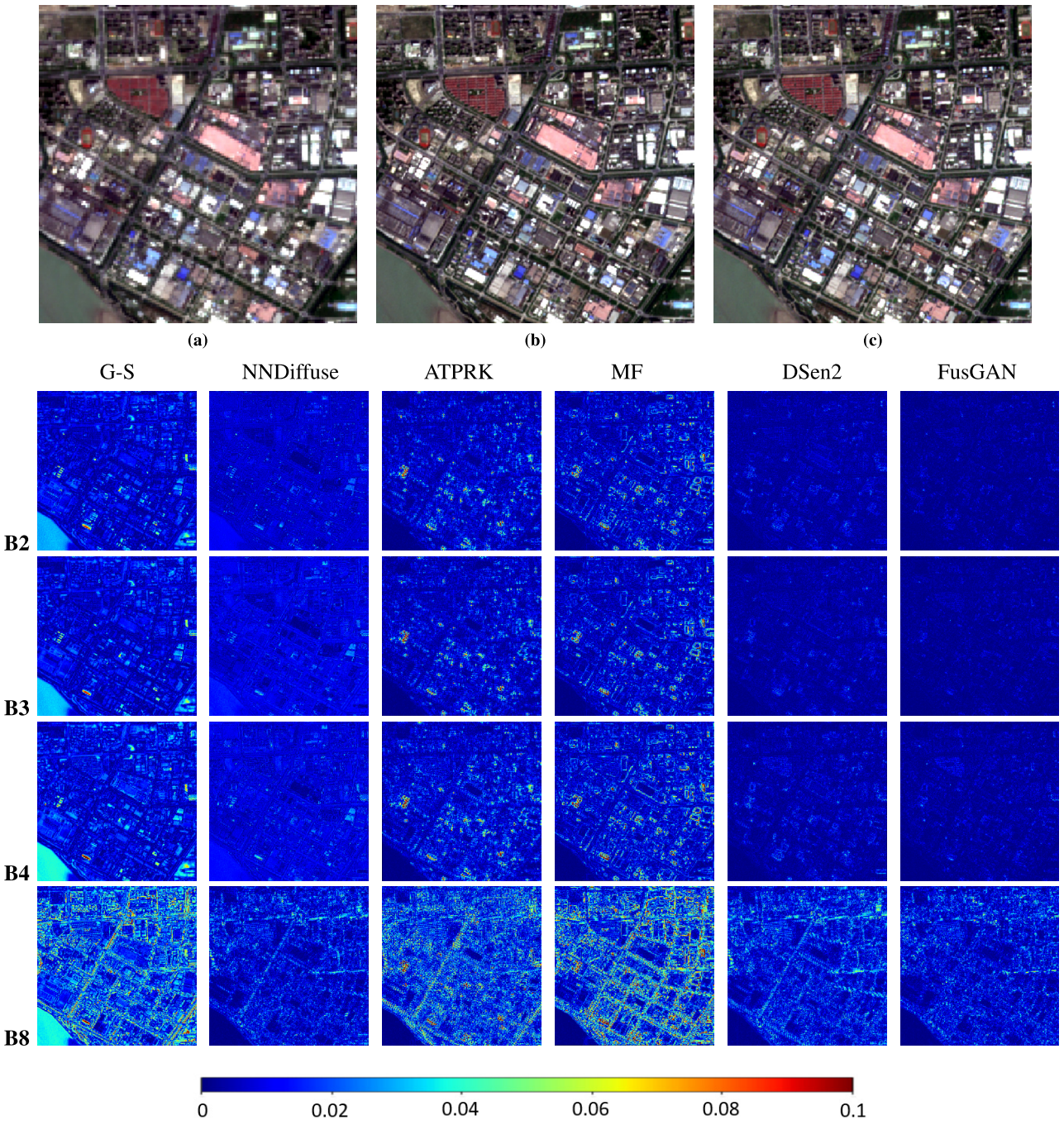


Fig. 6. Error maps for each spectral band of the full-spatial resolution fused images. The maps show absolute value differences ranging from 0 to 1. The top three images are displayed with a true-color composition (RGB: B4-B3-B2). (a) Original image. (b) Fused image by FusGAN. (c) Reference image.

band of B5 and the middle infrared band of B11 perform somewhat better than the other spectral bands. Moreover, two subset regions with respect to the full-resolution and full-spectral images are adopted for pixelwise error analysis. Specifically, the absolute differences between the fused and reference images are calculated to indicate the fusion errors of each pixel in the spectral bands. Fig. 6 shows that in full-spatial resolution image fusion, the errors in B2, B3, and B4 derived in the FusGAN approach are significantly lower than the errors derived in G-S, NNDiffuse, ATPRK, and MF and somewhat lower than the those derived in the DSen2 method. For the B8 band of the fused images, the FusGAN approach achieved significantly lower errors than those achieved in G-S, ATPRK, and MF and somewhat lower errors than those achieved in NNDiffuse and DSen2. In full-spectral image fusion (Fig. 6), the FusGAN approach attained remarkable improvements in all spectral bands by comparing the error maps obtained in the different testing methods.

*2) Visual Performance Evaluation of the Fused Satellite Image:* Due to the lack of ground truth superresolved LR images, only visual evaluation of the fusion of real images is performed in this section. In the case of Sentinel-2 image

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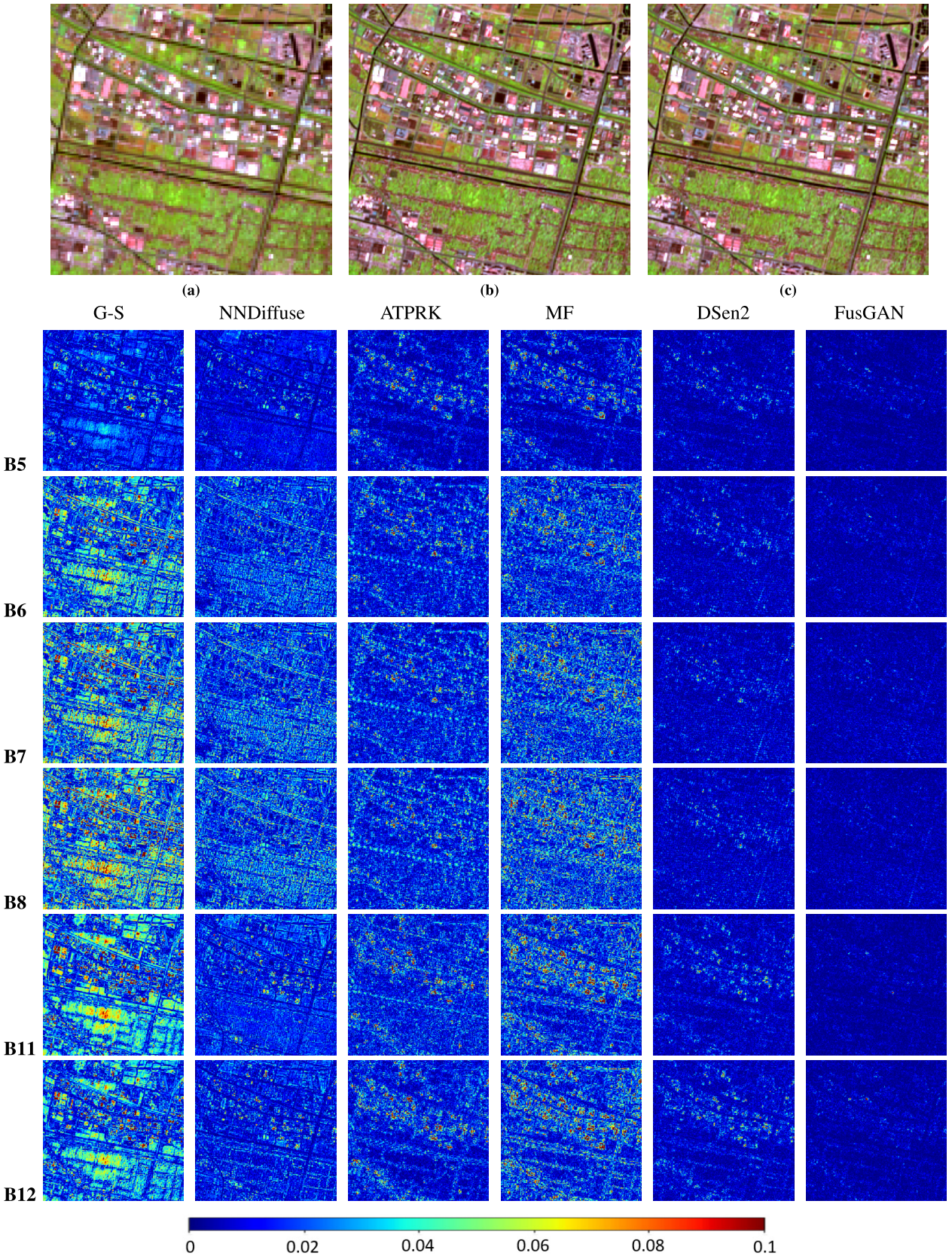


Fig. 7. Error maps for each spectral bands of the full-spectral fused images. The maps show absolute value differences ranging from 0 to 1. The top three images are displayed with a false-color composition (RGB: B11-B6-B5). (a) Original image. (b) Fused image by FusGAN. (c) Reference image.

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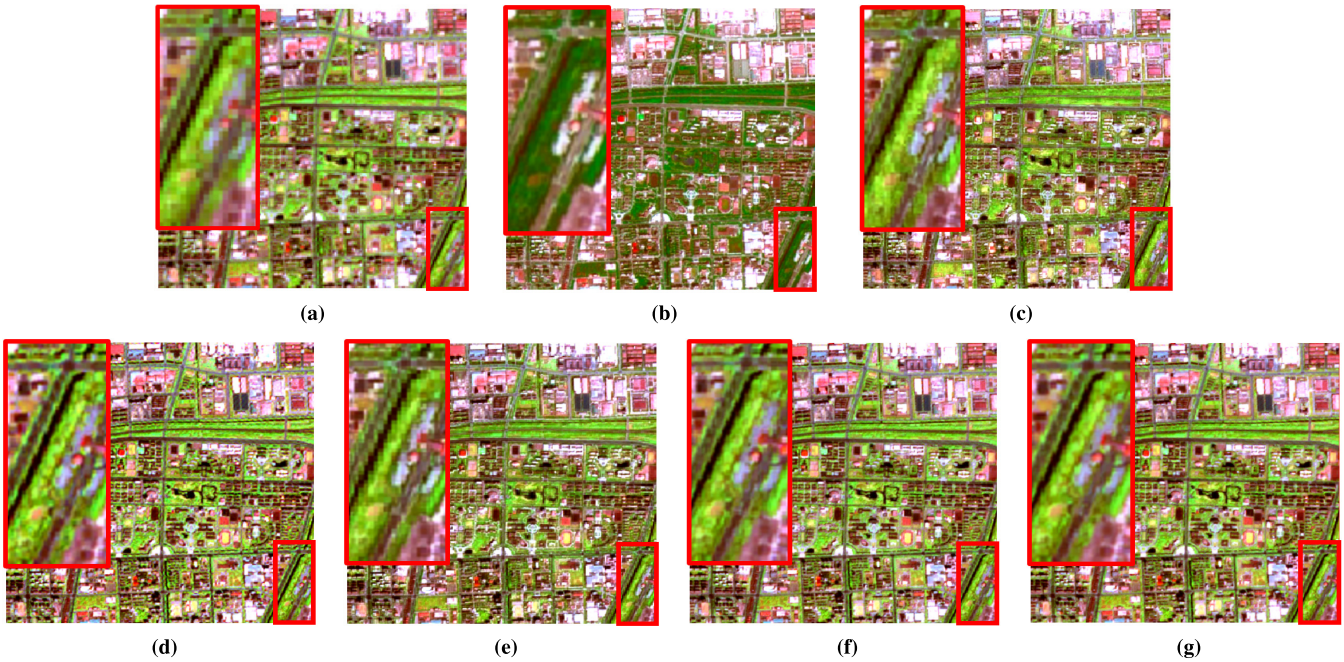


Fig. 8. Sentinel-2 image fusion results by the proposed and comparison methods. (a) Original image (LR bands). (b) Fused image by G-S. (c) Fused image by NNDiffuse. (d) Fused image by ATPRK. (e) Fused image by MF. (f) Fused image by DSen2. (g) Fused image by FusGAN.

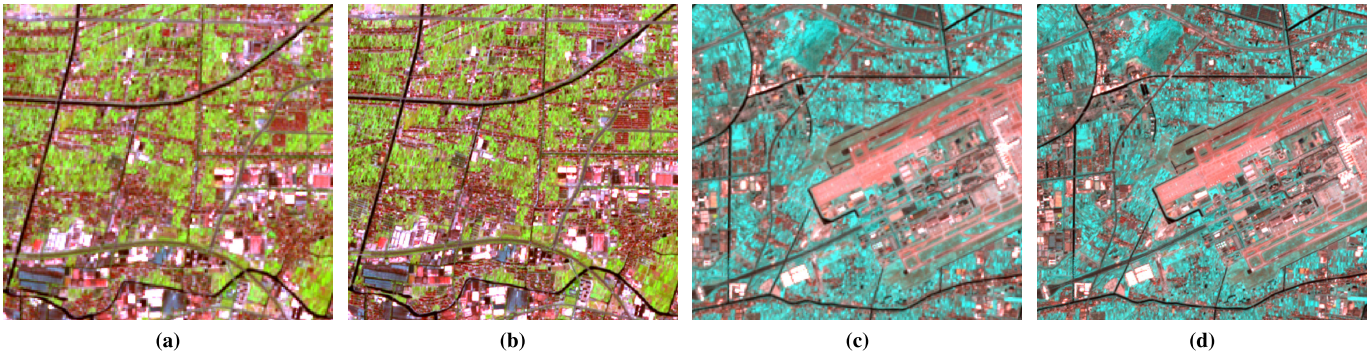


Fig. 9. FusGAN results of real Sentinel-2 image fusion in two scenes (bands 11, 6, and 5 and bands 12, 8a, and 7, respectively, are set as RGB). The left and right figures are the original and fused LR images, respectively. (a) Sentinel-2 image scene 1. (b) FusGAN result for Sentinel-2 image scene 1. (c) Sentinel-2 image scene 2. (d) FusGAN result for Sentinel-2 image scene 2.

fusion (Fig. 8), the G-S method produces serious errors in preserving radiation information [Fig. 8(b)], and the NNDiffuse and MF methods suffer from a certain degree of geometric distortion [Fig. 8(c) and (e)]. The DSen2 method performs better, while some geometric distortion can also be observed in the fused images [Fig. 8(f)]. In the case of Landsat 8 image fusion (Fig. 10), the proposed FusGAN approach achieved a remarkably better performance compared to that of the other methods considered. The G-S method has a decreased in reconstructing both the spatial and radiation information [Fig. 10(b)], and the NNDiffuse, ATPRK and MF methods perform well in detailed spatial information reconstruction; however, clear distortion can be observed in the fusion results [Fig. 10(c)–(e)]. Compared with the proposed FusGAN method, DSen2 obtained a similar accuracy, while some radiation information loss and spatial blurring can be observed in the fused images [Fig. 10(f)]. To better evaluate the visual performance of the FusGAN method, the results of two additional subregions of the test Sentinel-2 and Landsat OLI images are presented below. Clearly, as shown in Figs. 9 and 11, all the fused images obtained with the FusGAN method are notably improved compared with the original LR images.

*3) Parameter Sensitivity Analysis:* The effectiveness of the deep learning model depends on the appropriate selection of parameters. Therefore, we present the indispensable discussion of the sensitivity of key parameters involved in the proposed FusGAN method. These parameters include the patch size of the training data, the number of residual dense blocks, and the $\lambda$ parameter mentioned in (4). Some of them are discussed in the following.

1) The patch size of the HR image is set to 32 in the research by Lanaras *et al.* [54]. Additionally, we further test the FusGAN performance by setting the patch size at 48 and 64 in this study.

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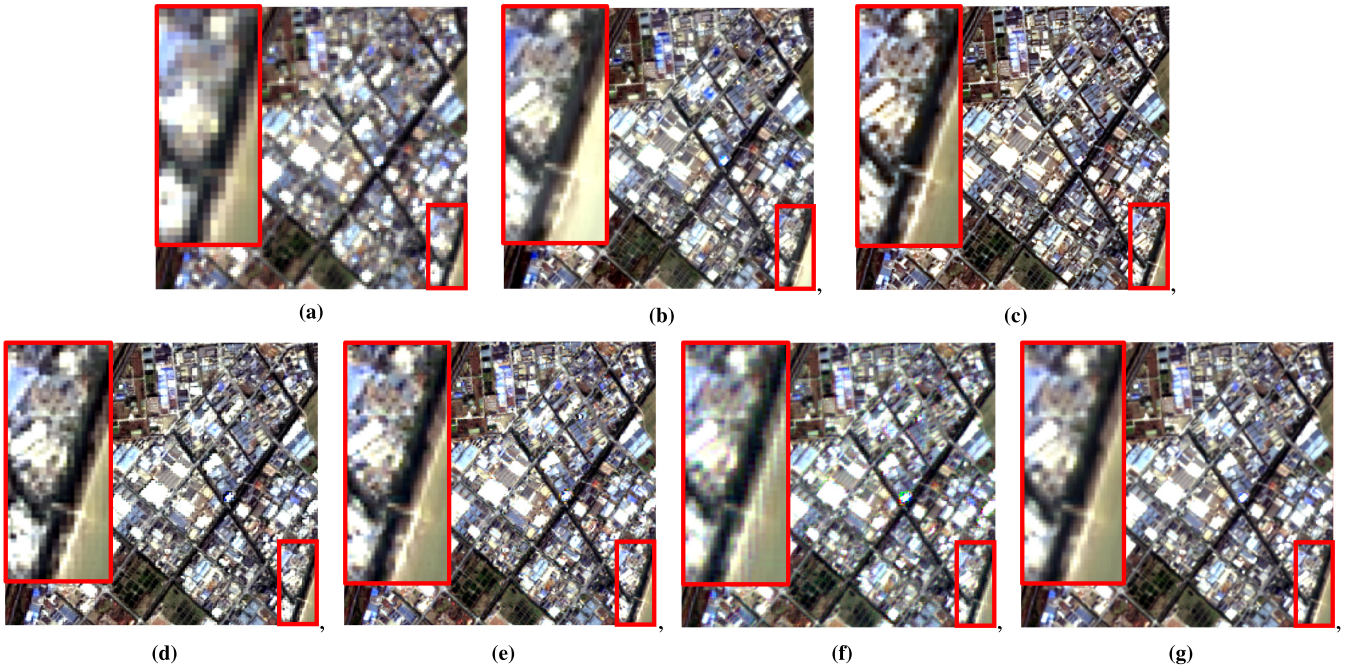


Fig. 10. Landsat 8 image fusion results by the FusGAN and comparison methods. (a) Original image (LR bands). (b) Fused image by G-S. (c) Fused image by NNDiffuse. (d) Fused image by ATPRK. (e) Fused image by MF. (f) Fused image by DSen2. (g) Fused image by FusGAN.

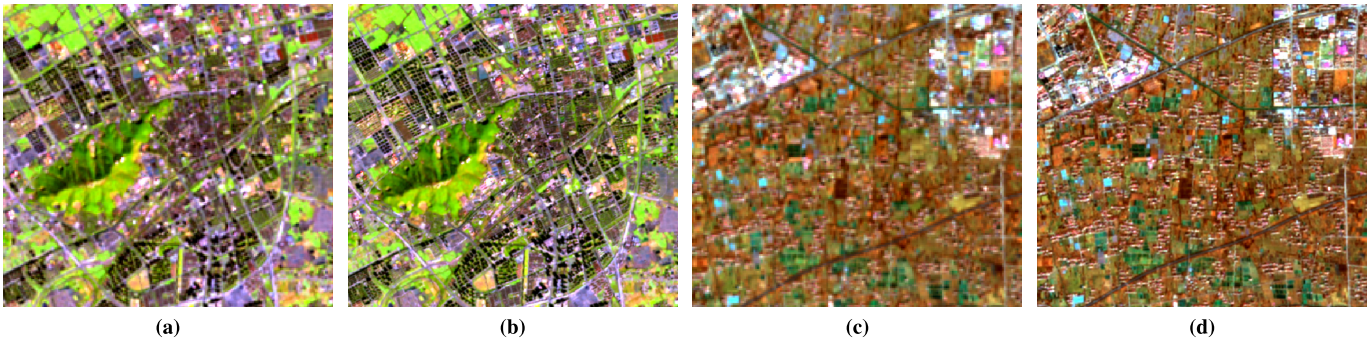


Fig. 11. FusGAN results of real Landsat image fusion in two additional subregions (bands 6, 5, and 4 and bands 7, 4, and 1, respectively, are set as RGB). (Left) original and (Right) fused LR images, respectively. (a) Landsat-8 image scene 1. (b) FusGAN result for Landsat-8 image scene 1. (c) Landsat-8 image scene 2. (d) FusGAN result for Landsat-8 image scene 2.

2) The numbers of the residential dense blocks are investigated from 2 to 6.
3) We follow the research by Ledig *et al.* [72] and set the $\lambda$ parameter to 1000. In addition, the image fusion performance is tested by setting $\lambda$ to 50, 100, 500, and 2000.

We experimentally analyze the effects of different parameter settings on the image fusion performance, and the SRE values corresponding to different parameters setting are shown in Fig. 12. Generally, we conclude that: 1) the fusion accuracy is the lowest when the patch size is set to 32. Through a comprehensive comparison of different accuracy metrics in terms of SRE, MAE, SSIM, and SAM, the FusGAN achieved similar outstanding performance when the patch size is set to 48 or 64; 2) the image fusion accuracy increases with the increase in the number of residual dense blocks from 2 to 4, and then decreases with the increase in the number of residual dense blocks from 4 to 6. With respect to time consumption, the running time of FusGAN linearly increases with an increase in the number of residual dense blocks; and 3) the SRE values vary very little (less than 0.11) among the different $\lambda$ settings. Specifically, the image fusion accuracy slightly increases with an increase in $\lambda$, and the image fusion accuracy becomes stable when $\lambda$ reaches 1000.

*4) Ablation Analysis:* Since the new proposed FusGAN method involves multiple state-of-the-art techniques, i.e., residual dense blocks and generative adversarial training, we investigate and analyze the performance improvement of our FusGAN method by step-wise adding the different components. In this study, the satellite image fusion was improved using a residual dense network connection and the GAN theory based on the traditional convolutional network; therefore, the ConvNet with traditional convolutional blocks is used as the baseline method in this analysis, and the ConvNet with residual blocks, ConvNet with residual dense blocks and GAN with residual dense blocks are implemented for the image fusion improvement analysis. As shown in Table III, the ConvNet-based methods performed differently with

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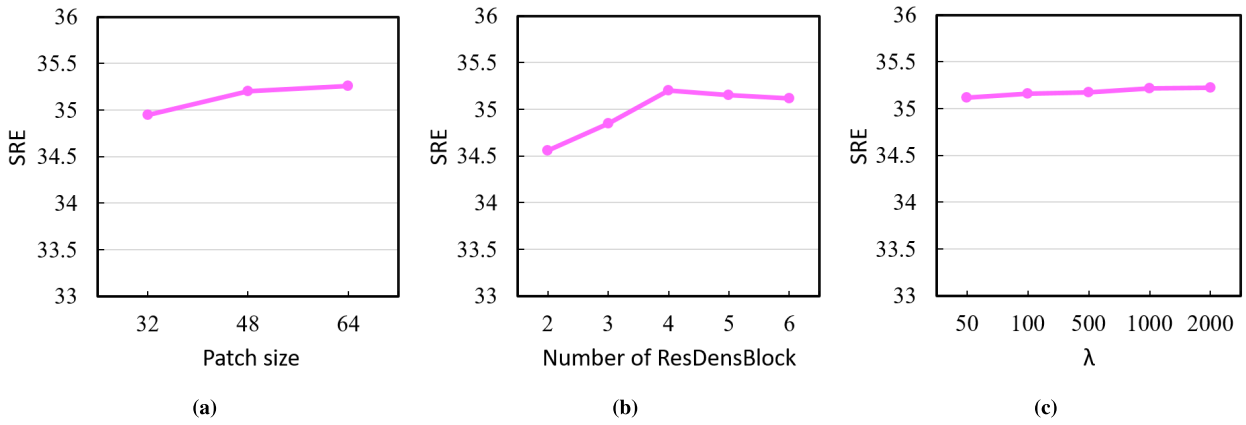


Fig. 12. Parameter sensitivity analysis of the proposed FusGAN model with respect to the (a) patch size, (b) number of residual dense blocks, and (c) $\lambda$.

TABLE III
ABLATION ANALYSIS OF THE PROPOSED FUSGAN MODEL

| | ConvNet-based | | | GAN-based |
|---|---|---|---|---|
| | ConvBlock | ResBlock | ResDensBlock | ResDensBlock |
| SRE | 33.941 | 34.742 (+0.781) | 34.904 (+0.162) | 35.202 (+0.298) |

different convolutional blocks; specifically, the residual block outperforms the traditional convolutional block, and the residual dense block outperforms the residual block. We compare the ConvNet-based and GAN-based methods with the same residual dense blocks in terms of image fusion accuracy, and the results indicate that the GAN-based method significantly outperforms the ConvNet-based method. As indicated above, the residual dense block and generative adversarial training of the proposed FusGAN model compose a reasonable and advanced design for the improvement of satellite image fusion.

*5) Computational Cost of the Proposed FusGAN Model:* All the experiments in this study were implemented in the Keras framework with TensorFlow as the back-end. The model training and inference are running on a Tesla K80 GPU, with 12 GB of RAM. We conduct our experimental analysis on the $2000 \times 2000$ Landsat-8 and Sentinel-2 images, respectively, and compare the computational cost among the proposed FusGAN, ResDensBlocks-based ConvNet and DSen2 methods. Specifically, in the model training stage, the GAN model training consumes approximately three times more computational time than the CNN model training; i.e., DSen2, ConvNet with ResDensBlock, and FusGAN consume 3.7, 5.7, and 16.9 s for one training epoch, respectively. The number of model parameters is consistent with the consumption of time; i.e., the FusGAN model has the most parameters (409, 167) which exceeds the number of parameters for the ResDensBlock-based ConvNet model (240, 364) and the DSen2 model (76, 324). As indicated above, training the GAN model requires more memory overhead than the ConvNet model training. In the model inference stage, since only the generative network of the GAN model is used for image fusion, the GAN model performs as efficiently as the ConvNet model (Table IV). The FusGAN method consumes more time than the exiting DSen2 method; however, the FusGAN method only consumes approximately 5 seconds for the processing of a $2000 \times 2000$ satellite image, and we think the computational cost is still acceptable in practice.

## IV. DISCUSSION

### *A. Key Issues of Deep Learning-Based Satellite Image Fusion*

The deep learning-based method captures the transfer relation between the multiresolution input and full HR output through learning from the example data; however, images are not available at full spatial and spectral resolutions, which complicates model training and performance assessment alike. The Wald protocol [68] consists of downscaling both the original HR and LR components so that the original LR component can be adopted as a reference for transfer relation learning and performance assessment. Based on the assumption of scale invariance, ConvNet was trained with reduced-resolution images and was used to fuse the original multiresolution images. However, the assumption of scale invariance lacks rigorous proof and inevitably leads to errors, particularly when the scale gap increases [54]. Another factor impacting the performance of image fusion is the ConvNet architecture. To accurately and efficiently capture the transfer relation between the multiresolution and full HR images, many researchers have employed residual connections to build deep ConvNets [48], [53], [54], which has resulted in better image fusion performance. In conclusion, both the scale invariance assumption and the less advanced network design would lead to errors in satellite image fusion. In this study, we introduced the adversarial training mechanism, which discriminates the fused images generated by the downscaled-image-trained G-Net, thereby suppressing the errors caused by the scale invariance assumption to a certain extent [Fig. 13(b)]. Moreover, we build the G-Net with state-of-the-art residual dense blocks, and based on comparative evaluation, these residual dense blocks are particularly suitable for satellite image fusion in comparison with the traditional convolutional

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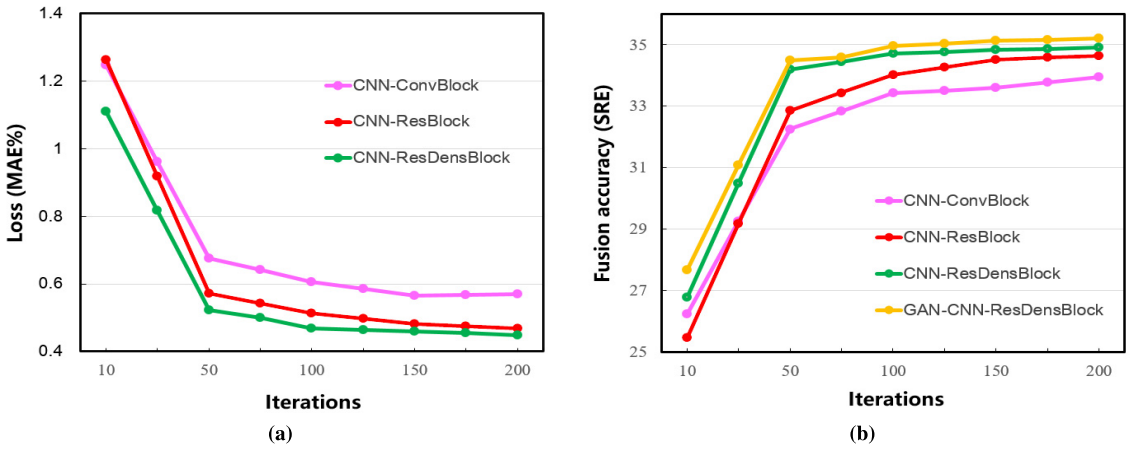


Fig. 13. Illustration of the improvement in terms of the model performance (model loss value) and fusion accuracy (Fusion accuracy SRE) by using (a) ResDensBlock and (b) adversarial training, respectively.

TABLE IV
PROCESSING TIME COMPARISONS AMONG THE DSEN2, RESDENSBLOCKS-BASED CONVNET AND PROPOSED FUSGAN METHODS ON THE SATELLITE IMAGE FUSION

| Data/Time consuming (s) | DSen2 | ResDensBlocks-based ConvNet | FusGAN |
|---|---|---|---|
| Landsat-8 | 1.6 | 5.86 | 5.79 |
| Sentinel-2 | 1.35 | 5.64 | 5.65 |

block and classical residual block [Fig. 13(a)]. By means of the improvements in both the training strategy and network architecture, the proposed FusGAN approach outperforms the state-of-the-art deep learning-based method (Fig. 13).

### *B. Potential of GAN-Based Satellite Image Fusion*

The proposed FusGAN method in our study performs the best in remote sensing image fusion not only compared to the classical G-S and NDDiffuse methods but also compared to the state-of-the-art ATPRK and MF methods and the deep learning-based DSen2 method. The main reason for the improvement is that we introduced the GAN theory in the image fusion objective. Compared with ConvNet, the GAN is particularly suitable for solving image reconstruction problems due to the adversarial training mechanism, which lends it an innate advantage in satellite image fusion. The GAN contains two ConvNet models, namely, the generative and discriminative networks. To our knowledge, few GAN-based remote sensing image fusion methods have been reported in the literature, and we conduct an early examination of GAN-based image fusion in this study. Particularly, with the customized training strategy and neural network structure, the FusGAN method is proposed for improved satellite image fusion. Nevertheless, the GAN-based method in remote sensing image fusion can still be greatly improved, and therefore, additional research is required in future work.

## V. CONCLUSION

In this article, we developed a new FusGAN approach for remote sensing image fusion. Generally, the downscaled-image-trained G-Net is used for fusion of original images, and the fused pairwise images are discriminated by the D-Net. Through generative adversarial training between the G-Net and D-Net, the capability of the structured G-Net is constantly improved during training. Moreover, we adopt residual dense blocks to build the G-Net and thus make the feature learning process of the G-Net more efficient and accurate. The proposed FusGAN approach is quantitatively and qualitatively evaluated by using simulated and real satellite images, respectively. As the results demonstrate, the FusGAN approach performs better than the existing image fusion methods, including the classical G-S and NNDiffuse methods, the state-of-the-art ATPRK and MF methods, and the newly proposed deep learning-based DSen2 method. Furthermore, the proposed FusGAN method can be applied to remote sensing images with different spatial resolutions and combinations of HR and LR images to realize high-fidelity image fusion.

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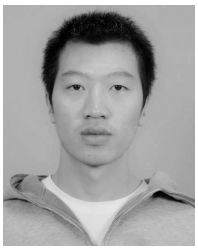

**Xin Luo** (Member, IEEE) received the B.S. degree in land resource management from Sichuan Agricultural University, Chengdu, China, in 2012, and the M.E. and Ph.D. degrees in photogrammetry and remote sensing from Tongji University, Shanghai, China, in 2015 and 2019, respectively.

He is a Post-Doctoral Fellow with the MNR Key Laboratory for Geo-Environmental Monitoring of Coastal Zone, Ministry of Natural Resources Great Bay Area & Guangdong Key Laboratory of Urban Informatics & Shenzhen Key Laboratory of Spatial Smart Sensing and Services, Shenzhen University, Shenzhen, China. His research interests include multispectral/hyperspectral remote sensing data processing, deep learning, and urban remote sensing application.

**Xiaohua Tong** (Senior Member, IEEE) received the Ph.D. degree from Tongji University, Shanghai, China, in 1999.

From 2001 to 2003, he was a Post-Doctoral Researcher with the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing, Wuhan University, Wuhan, China. He was a Research Fellow with Hong Kong Polytechnic University, Hong Kong, in 2006, and a Visiting Scholar with the University of California at Santa Barbara, Santa Barbara, CA, USA, from 2008 to 2009. His research interests include photogrammetry and remote sensing, trust in spatial data, and image processing for high-resolution satellite images.

**Zhongwen Hu** (Member, IEEE), photograph and biography not available at the time of publication.

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